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C2 — Chemical Patterns

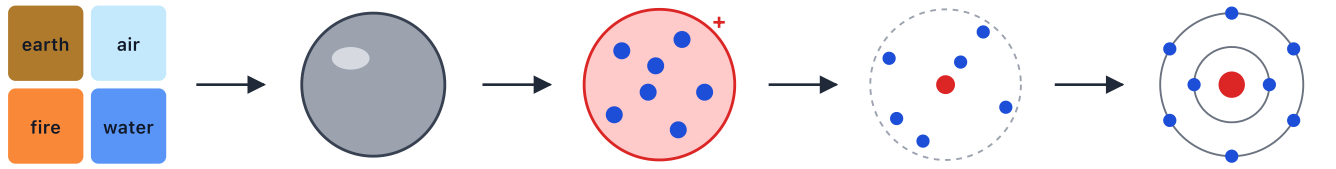
OCR GCSE Chemistry B (J258) — Twenty First Century Science · All papers (J258/01 & J258/03 Breadth in Chemistry, J258/02 & J258/04 Depth in Chemistry)

There is no Higher-tier-only **(HT)** content in C2: every statement in this chapter can be examined on both Foundation and Higher tiers.

C2.1 How Have Our Ideas About Atoms Developed Over Time?

The modern model of the atom was built up step by step. Each model was **rejected, modified or extended** when new evidence became available that it could not explain.

Development of the Atomic Model



Ancient Greeks
c. 400 BC
four elements: earth, air, fire, water

Dalton
1803
tiny solid spheres; cannot be divided

Thomson
1897
"plum pudding": electrons in a ball of + charge

Rutherford
1911
tiny dense + nucleus; mostly empty space

Bohr
1913
electrons in fixed shells (energy levels)

electrons shown in blue, positive charge / nucleus in red (not to scale)

Scientist / date	Model	Evidence / reason for change

Ancient Greeks (c. 400 BC)	Everything made of four "elements": earth, air, fire and water. (Democritus also suggested matter was made of tiny indivisible particles — *atomos*.)	Ideas based on reasoning, not experiment.
John Dalton (1803)	First particle model: elements are made of tiny, solid, **indivisible atoms**; atoms of the same element are identical; atoms combine in fixed ratios to form compounds.	Explained his measurements of the masses of elements that combine in reactions.
J.J. Thomson (1897)	**"Plum pudding" model**: a ball of positive charge with negative **electrons** embedded in it.	He discovered the electron (cathode-ray experiments) — so atoms are *not* indivisible and contain smaller charged particles.
Ernest Rutherford (1911)	**Nuclear model**: a tiny, dense, positive **nucleus** containing most of the mass, with electrons around it; most of the atom is empty space.	Geiger and Marsden's **gold foil (alpha scattering) experiment**: most alpha particles passed straight through, but a few were deflected through large angles — impossible with a plum pudding.
Niels Bohr (1913)	Electrons orbit the nucleus in fixed **shells** (energy levels) at set distances.	Explained why atoms emit light only at particular wavelengths (line spectra), and why atoms are stable. His calculations agreed with experimental observations.

Later, the **proton** (Rutherford, about 1919) and the **neutron** (James Chadwick, 1932) were discovered inside the nucleus.

> **Ideas about Science (IaS3) — how models develop.** Scientists suggested explanations, made predictions from them, checked those predictions by experiment, and built on each other's work (Rutherford was Thomson's student; Bohr worked with Rutherford). A model is kept until new evidence it cannot explain appears; then it is modified or replaced. Creative

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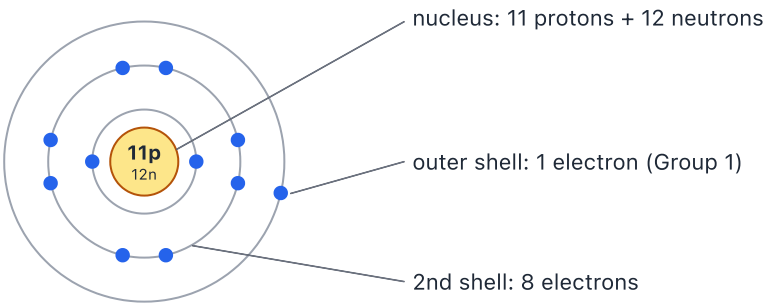
> thinking is needed to propose a new model — it does not "emerge" automatically from the data.

The Structure of the Atom

An atom has a small, **positively charged nucleus** (protons and neutrons) surrounded by **negatively charged electrons** in shells. The radius of the nucleus is much smaller than the radius of the atom, but **almost all of the mass is in the nucleus**. Atoms are neutral overall: number of electrons = number of protons.

Particle	Relative charge	Relative mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small (about 1/2000, often written ≈ 0)	shells around the nucleus

Bohr Model of a Sodium Atom ($^{23}_{11}$Na)



Electron arrangement 2.8.1 → Period 3 (3 shells)

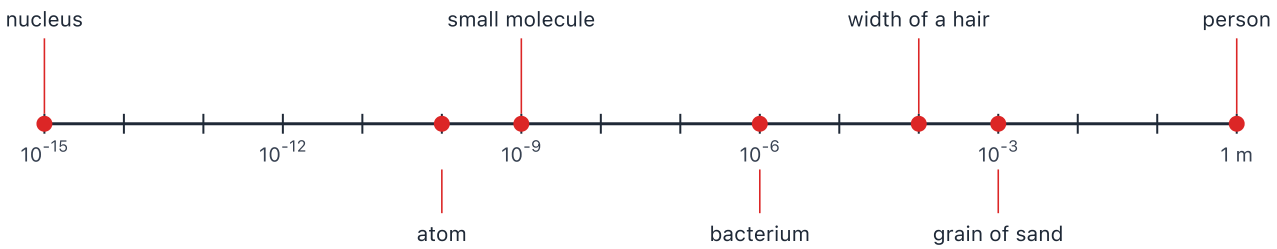
Size and Scale

- Atoms are about **10^{-10} m** across (0.1 nm).
- The nucleus is about one hundred-thousandth (1/100 000) of the diameter of the atom: about **10^{-15} m**.

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- Small molecules (e.g. H_2O, CO_2, CH_4) are about **10^{-10} to 10^{-9} m**; molecules contain from two to hundreds of atoms (large molecules such as proteins and polymers are much bigger).
- Anything you can see with the naked eye contains many millions of atoms.

Size and Scale (logarithmic scale: each tick is ×10)



Atom ≈ 10^{-10} m · nucleus ≈ 10^{-15} m (100 000 times smaller) · small molecules ≈ 10^{-9} m

Worked Example: A human hair is about 1×10^{-4} m wide. Roughly how many atoms (diameter 1×10^{-10} m) would fit side by side across it?
number = $(1 \times 10^{-4}) \div (1 \times 10^{-10}) = 1 \times 10^{-4+10} =$ **1×10^{6}** — about a million atoms.

Worked Example: If an atom were scaled up to the size of a football stadium (about 100 m across), how big would the nucleus be?
nucleus = atom ÷ 100 000 = 100 m ÷ 10^5 = 10^{-3} m = **1 mm** — about the size of a grain of sand in the centre circle. The atom is mostly empty space.

Atomic Number, Mass Number and Isotopes

Atomic number (Z) = number of protons (the order of the element in the Periodic Table).
Mass number (A) = number of protons + number of neutrons.
Isotopes are atoms of the same element (same number of protons) with different numbers of neutrons, so different mass numbers.

protons = atomic number · electrons = protons (in an atom) · neutrons = mass number − atomic number

Atom	Atomic number	Mass number	Protons	Neutrons	Electrons
$^{12}_{6}C$	6	12	6	6	6
$^{14}_{6}C$ (isotope)	6	14	6	8	6
$^{23}_{11}Na$	11	23	11	12	11
$^{35}_{17}Cl$	17	35	17	18	17
$^{37}_{17}Cl$ (isotope)	17	37	17	20	17

Exam Tip: The Periodic Table gives the **relative atomic mass** (e.g. 35.5 for chlorine), which is an average over the isotopes. When asked about a specific atom, use its **mass number**; if you are told to use the Periodic Table, round the relative atomic mass to the nearest whole number (e.g. Na 23.0 → 12 neutrons).

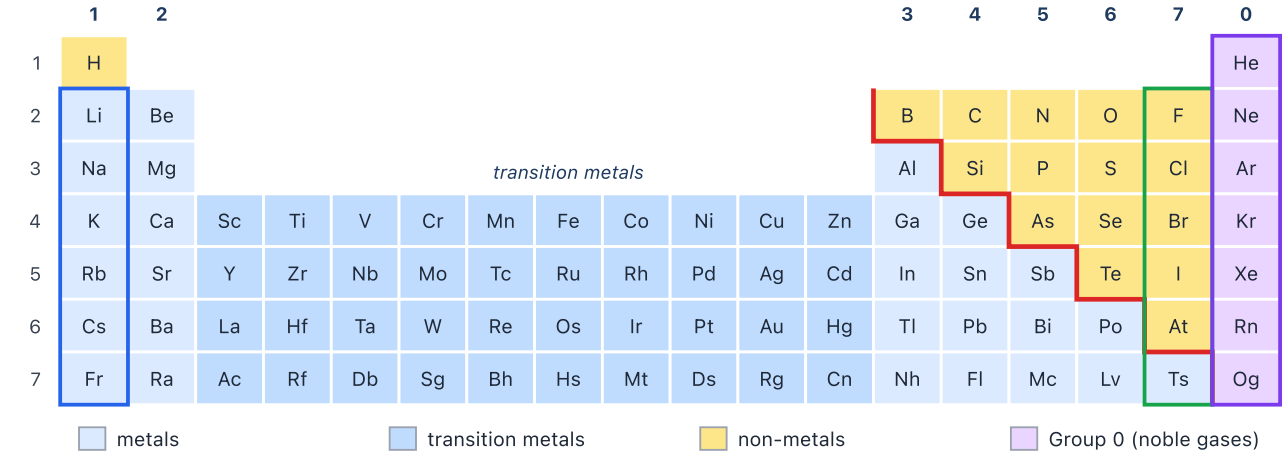
C2.2 What Does the Periodic Table Tell Us About the Elements?

Structure of the Modern Periodic Table

- Elements are arranged in order of **atomic number** (number of protons).
- **Groups** are the vertical columns. Elements in the same group have the **same number of electrons in their outer shell** (Group 1 = 1, Group 7 = 7, Group 0 = full outer shell) and so similar chemical properties.
- **Periods** are the horizontal rows. The period number = **number of occupied electron shells**. The number of shells increases down a group and stays the same across a period.

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The Modern Periodic Table (outline)



Red staircase: approximate metal / non-metal boundary. Boxes: Group 1 (blue), Group 7 (green), Group 0 (purple). Rows = periods, columns = groups.

Worked Example: Silicon has atomic number 14. Where is it in the Periodic Table?

14 electrons: arrangement **2.8.4** → 3 shells = **Period 3**; 4 outer electrons = **Group 4**. The position follows from the electron arrangement, which follows from the atomic number.

Electron Arrangements of the First 20 Elements

Fill shells from the nucleus outwards: up to **2** in the first shell, then up to **8** in the second and **8** in the third (for the first 20 elements), then the fourth.

Element	Z	Arrangement	Element	Z	Arrangement
H	1	1	Na	11	2.8.1
He	2	2	Mg	12	2.8.2
Li	3	2.1	Al	13	2.8.3
Be	4	2.2	Si	14	2.8.4
B	5	2.3	P	15	2.8.5
C	6	2.4	S	16	2.8.6
N	7	2.5	Cl	17	2.8.7

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O	8	2.6	Ar	18	2.8.8
F	9	2.7	K	19	2.8.8.1
Ne	10	2.8	Ca	20	2.8.8.2

Mendeleev's Periodic Table

In 1869 **Dmitri Mendeleev** arranged the known elements in order of **relative atomic mass**, and put elements with **similar properties** (e.g. the formulae of their oxides and chlorides) in the same columns. He did not know about protons or electrons.

- He **left gaps** for elements not yet discovered, and **predicted their properties** from the elements around the gaps.
- He **swapped the order of some elements** (e.g. tellurium and iodine) so that they fitted with elements of similar properties, even though this broke the atomic mass order.

Property	Mendeleev's prediction for "eka-silicon" (1871)	Germanium (discovered 1886)
Relative atomic mass	72	72.6
Density / g cm^{-3}	5.5	5.3
Formula of oxide	EsO_2	GeO_2
Appearance	grey metal	grey-white metalloid

Why his ideas were accepted: when new elements (gallium, scandium, germanium) were discovered, they **fitted his gaps** and had the properties he predicted. Later, the discovery of protons showed that ordering by **atomic number** (not mass) puts tellurium (Z = 52) before iodine (Z = 53) — exactly as Mendeleev had done. Atomic structure *explained* the order he proposed: the reversed pairs occur because of isotopes (tellurium's common isotopes are heavier).

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Ideas about Science (IaS3 and IaS1). Mendeleev's table is an example of an explanation that needed a *leap of imagination*. It was accepted because it made **predictions that were tested and found to be correct**, and because a later model (atomic structure) supported it. You should also be able to make and test predictions about trends and patterns in the Periodic Table (IaS1).

Metals and Non-Metals

Metals are on the **left** and in the **centre** of the Periodic Table; non-metals are on the **right** (above the staircase line). Metal atoms have few outer electrons (1, 2 or 3) which they lose in reactions; non-metal atoms have more outer electrons (4 to 7) and tend to gain or share electrons.

Property	Metals	Non-metals
Melting and boiling points	Usually high	Usually low (many are gases)
State at room temperature	Solids (except mercury)	Gases, a liquid (bromine), or solids
Appearance	Shiny (lustrous) when freshly cut or polished	Dull (solids); various colours
Density	Usually high	Usually low
Electrical (and thermal) conductivity	Good conductors	Poor conductors / insulators (except graphite)
Other physical properties	Malleable and ductile	Brittle when solid
Ions formed	Positive ions (e.g. Na^+, Mg^{2+})	Negative ions (e.g. Cl^-, O^{2-})
Oxides	Basic, e.g. Na_2O, MgO (ionic)	Acidic, e.g. CO_2, SO_2 (covalent)
Relative reactivity	Most reactive at the bottom left (Group 1)	Most reactive at the top right (fluorine)

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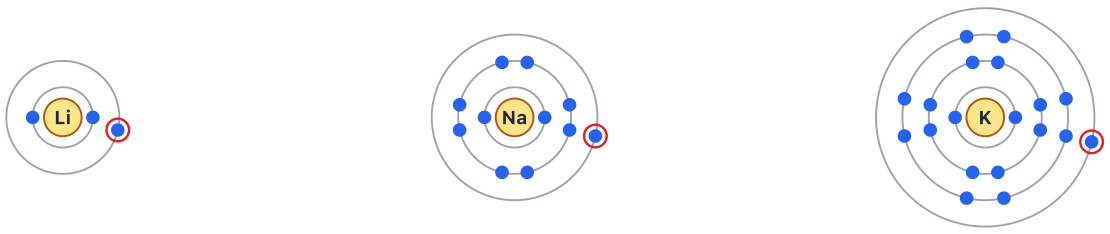
Group 1 — the Alkali Metals

Lithium, sodium, potassium, rubidium, caesium. All have **1 electron in the outer shell**. They are soft metals that can be cut with a knife, have **low densities** (Li, Na and K float on water) and **low melting points** for metals. They are stored under oil because they react with air and water.

Reaction	Observations	Equation (sodium)
Moist air	Shiny freshly cut surface quickly **tarnishes** (goes dull) as an oxide/hydroxide layer forms; faster down the group	$4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$
Water	Floats, fizzes (hydrogen), moves about, melts into a ball (Na); K gives a lilac flame. Solution becomes **alkaline** (universal indicator turns purple)	$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$
Chlorine	Heated metal burns brightly in chlorine (Na: yellow-orange flame), forming a white solid chloride	$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$

Metal	With water
Lithium	Fizzes steadily, floats, slowly disappears
Sodium	Fizzes rapidly, melts into a silvery ball, whizzes across the surface
Potassium	Very vigorous; hydrogen ignites with a lilac flame; may spit or crackle

Why Group 1 Gets More Reactive Down the Group



Li 2.1 Na 2.8.1 K 2.8.8.1

More shells → outer electron (circled) further from the nucleus and more shielded → weaker attraction → lost more easily

Reactivity increases down Group 1 because the atoms get larger: the single outer electron is further from the nucleus and more shielded by inner shells, so it is **attracted less strongly** and is

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lost more easily.

Experiments: Group 1 Reactivity (Demonstration)

Group 1 reactions with water are carried out by the teacher as a **demonstration**: cut a rice-grain-sized piece of the metal with a scalpel on a tile, dry off the oil, drop it into a large trough of water behind a **safety screen**; wear goggles. Compare how vigorously Li, Na and K react (speed of fizzing, movement, flame), and test the solution with universal indicator. Keep pieces the same size so it is a fair comparison.

Group 7 — the Halogens

Fluorine, chlorine, bromine, iodine, astatine. All have **7 electrons in the outer shell** and exist as **diatomic molecules** (F_2, Cl_2, Br_2, I_2). They are non-metals, toxic, and their vapours are coloured.

Halogen	State at room temperature and pressure	Colour of element	Colour as a gas
Fluorine, F_2	gas	pale yellow	pale yellow
Chlorine, Cl_2	gas	pale green (yellow-green)	pale green
Bromine, Br_2	liquid	dark red-brown	orange-brown
Iodine, I_2	solid	grey-black (shiny)	purple

Down the group: melting and boiling points **increase** and colours get **darker**.

Reactions with Group 1 metals form white ionic salts called metal **halides**:

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s) \quad 2K(s) + Br_2(l) \rightarrow 2KBr(s) \quad 2Li(s) + I_2(s) \rightarrow 2LiI(s)$$

Displacement Reactions (PAG1 — Reactivity Trend)

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A **more reactive halogen displaces a less reactive halogen** from a solution of its metal halide. Add a few drops of halogen solution (chlorine water, bromine water, iodine solution) to potassium chloride, bromide and iodide solutions in a spotting tile or test tubes and look for colour changes.

Halogen added ↓ / Salt solution →	KCl (colourless)	KBr (colourless)	KI (colourless)
Chlorine water	—	turns **orange** (Br_2 formed)	turns **brown** (I_2 formed)
Bromine water	no reaction	—	turns **brown** (I_2 formed)
Iodine solution	no reaction	no reaction	—

$$Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$$
$$Cl_2(aq) + 2KI(aq) \rightarrow 2KCl(aq) + I_2(aq)$$
$$Br_2(aq) + 2KI(aq) \rightarrow 2KBr(aq) + I_2(aq)$$

This shows the order of reactivity: **chlorine > bromine > iodine**. **Reactivity decreases down Group 7** because the atoms get larger: the outer shell is further from the nucleus and more shielded, so an incoming electron is **attracted less strongly** and is **gained less easily**.

> **Exam Tip:** Group 1 and Group 7 trends go in **opposite** directions. Group 1: reactivity *increases* down (outer electron *lost* more easily). Group 7: reactivity *decreases* down (electron *gained* less easily). In both, the reason is the distance of the outer shell from the nucleus and shielding.

Predicting Reactions from the Periodic Table

- Elements in the same group react in similar ways and form compounds with similar formulae (e.g. LiCl, NaCl, KCl; NaF, NaCl, NaBr).
- Use the trend to predict reactivity: rubidium and caesium will react with water even more violently than potassium; fluorine is the most reactive halogen and astatine the least.

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- Predict physical properties by extending the trend: e.g. astatine is expected to be a dark (black) solid at room temperature, with a higher melting point than iodine.

> **Worked Example:** Predict the products and write the equation for rubidium reacting with water.
> Rubidium is in Group 1 like sodium, so it reacts the same way but more vigorously: **$2Rb(s) + 2H_2O(l) \rightarrow 2RbOH(aq) + H_2(g)$**. The solution is alkaline (rubidium hydroxide).

C2.3 How Do Metals and Non-Metals Combine to Form Compounds?

Group 0 — the Noble Gases

Helium, neon, argon, krypton, xenon, radon. Their atoms have a **full outer shell** of electrons (2 for helium, 8 for the others). This stable arrangement means they have **no tendency to lose, gain or share electrons**, so they are **unreactive (inert)**.

- They exist as **single atoms** (monatomic) with very weak forces between them, so they have very **low melting and boiling points** and are all **colourless gases** at room temperature and pressure.
- Boiling points **increase down the group** (He −269 °C, Ne −246 °C, Ar −186 °C, Kr −153 °C, Xe −108 °C) as the atoms get larger.
- Uses come from their lack of reactivity: argon in light bulbs and welding, helium in balloons (low density, non-flammable).

Outer Electrons Decide the Properties

- Elements react to get a **full outer shell** (like a noble gas). Metals with 1–3 outer electrons **lose** them; non-metals with 5–7 outer electrons **gain** electrons.
- So the number of outer electrons (and hence the group) determines how an element reacts, and the number of shells (the period) affects how easily this happens.
- Since the electron arrangement is fixed by the atomic number, the **atomic number determines both the position in the Periodic Table and the chemical reactions**.

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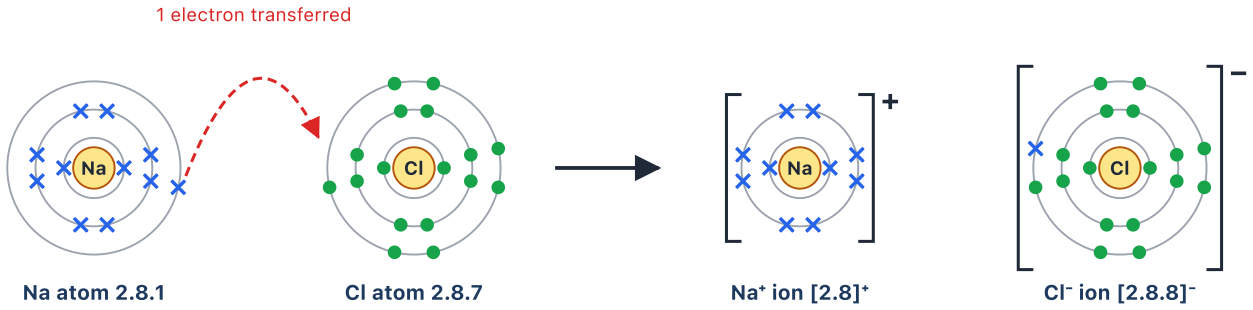
- Metals are on the left because their atoms have few outer electrons; non-metals on the right because they have nearly full outer shells.

Ionic Bonding

> An **ion** is an atom (or group of atoms) that has **lost or gained electrons** and so has an overall charge. **Ionic bonding** is the strong **electrostatic attraction between oppositely charged ions**. It forms when a metal reacts with a non-metal: electrons are **transferred** from the metal atoms to the non-metal atoms.

- Metal atoms lose their outer electrons to form **positive ions** with complete outer shells, e.g. Na (2.8.1) → Na^+ (2.8) + e^-.
- Non-metal atoms gain electrons to fill their outer shell, forming **negative ions**, e.g. Cl (2.8.7) + e^- → Cl^- (2.8.8).
- The **number of electrons lost or gained = the charge** on the ion: Group 1 → 1+, Group 2 → 2+, Group 6 → 2–, Group 7 → 1–.

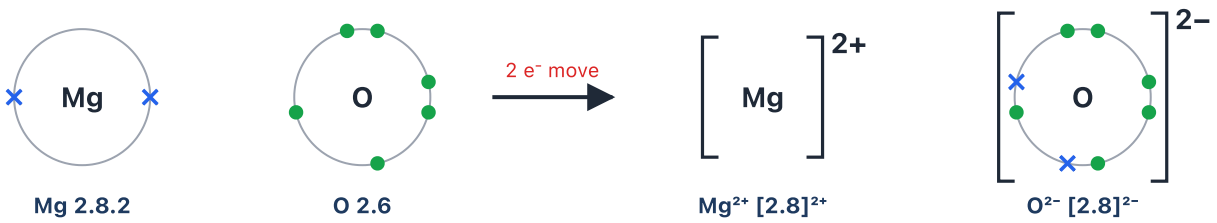
Dot and Cross Diagram: Forming Sodium Chloride



× = electron from sodium • = electron from chlorine. Both ions now have a full outer shell (like a Group 0 atom).
The oppositely charged ions attract by strong electrostatic forces = ionic bond.

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Outer-Shell Dot and Cross Diagram: Magnesium Oxide



Only outer shells shown. Mg^{2+} has lost its whole 3rd shell, so its outer (2nd) shell of 8 is usually not drawn.

Protons, Neutrons and Electrons in Ions

Ions have the same number of protons and neutrons as the atom; only the number of **electrons** changes. Positive ion: electrons = protons − charge. Negative ion: electrons = protons + charge.

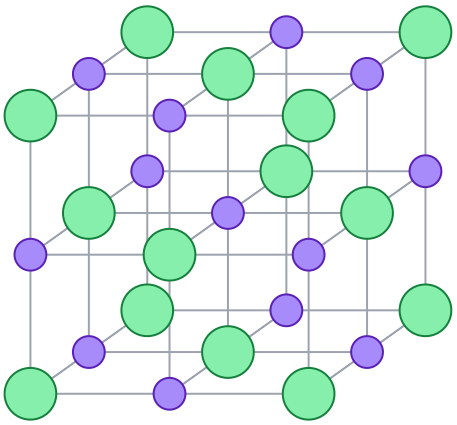
Particle	Protons	Neutrons	Electrons	Electron arrangement
$^{23}_{11}Na^{+}$	11	12	10	2.8
$^{35}_{17}Cl^{-}$	17	18	18	2.8.8
$^{24}_{12}Mg^{2+}$	12	12	10	2.8
$^{16}_{8}O^{2-}$	8	8	10	2.8
$^{39}_{19}K^{+}$	19	20	18	2.8.8

Giant Ionic Lattices

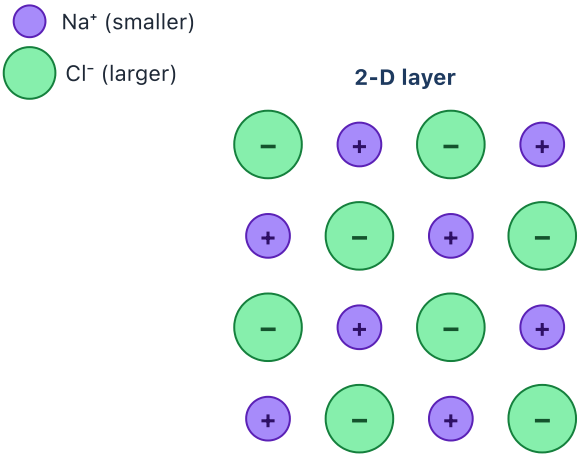
In an ionic compound such as sodium chloride, huge numbers of positive and negative ions are arranged in a regular, repeating three-dimensional **giant ionic lattice**. Each ion is surrounded by ions of the opposite charge and the electrostatic forces act **in all directions**.

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Sodium Chloride: 3-D and 2-D Representations of the Giant Ionic Lattice



3-D: each ion has 6 oppositely charged neighbours



ratio $Na^+ : Cl^- = 1 : 1$ → formula NaCl

Worked Example: In the 2-D layer above there are 16 ions: 8 Cl^- and 8 Na^+. What does this tell you about the formula?

ratio $Na^+ : Cl^- = 8 : 8 = 1 : 1$, so the formula is **NaCl**. The charges balance: (+1) + (−1) = 0. In the 3-D drawing, each Na^+ touches 6 Cl^- and each Cl^- touches 6 Na^+.

Properties of Ionic Compounds

Property	Explanation
High melting and boiling points (e.g. NaCl melts at 801 °C)	The strong electrostatic attraction between oppositely charged ions acts throughout the giant lattice; a **large amount of energy** must be transferred to overcome it.
Many dissolve in water	The charged ions interact with (are attracted to) water molecules, which pull them out of the lattice.
Conduct electricity when molten or dissolved	The ions are free to **move** and carry charge.
Do not conduct when solid	The ions are held in **fixed positions** in the lattice and cannot move.

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Hard but brittle crystals	The lattice is held strongly, but if layers shift, like charges line up and repel.

Substance	Bonding	Melting point / °C
Magnesium oxide, MgO	ionic (2+ and 2– ions: even stronger attraction)	2852
Sodium chloride, NaCl	ionic	801
Sodium, Na	metallic	98
Water, H_2O	small covalent molecules (weak forces between molecules)	0
Oxygen, O_2	small covalent molecules	–218

Practical: testing the properties of ionic compounds. Compare samples of, e.g., sodium chloride, potassium iodide and magnesium sulfate with a covalent substance such as wax or sugar: test solubility in water, test the electrical conductivity of the solid and of the solution with a simple circuit (battery, bulb or ammeter, carbon electrodes), and compare how easily they melt when heated.

Limitations of Models of Ions and Ionic Compounds

Model	Useful because it shows...	Limitations
Dot and cross diagram	Where the electrons come from and how they are transferred; the charges and full outer shells of the ions	Electrons are all identical (dots and crosses only show origin); does not show the 3-D lattice, the relative sizes of ions or that electrons are not in fixed positions; suggests one pair of ions rather than a giant structure

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2-D lattice drawing	The alternating arrangement and the ratio of ions	Only one layer; does not show the arrangement in space
3-D ball-and-stick model	The arrangement in space and the number of neighbours (6)	The "sticks" are not bonds — ions touch and the forces act in all directions; ions are far bigger relative to the gaps than drawn; not to scale
3-D space-filling model	The relative sizes of the ions and how closely they pack	Hard to see the inner ions and the arrangement; does not show the forces between ions

None of these models fully show the **movement** of ions (they vibrate), the nature of the electrons, or the interactions between the ions.

> **Ideas about Science (IaS3) — representational models.** Dot and cross diagrams and 2-D/3-D lattice drawings are *representational models*: they help us visualise and explain bonding and make predictions (e.g. formulae), but each simplifies reality. In an exam, be ready to say what a given model shows well **and** what it leaves out, and to translate between a 3-D picture, a 2-D drawing and a formula or ratio.

C2.4 How Are Equations Used to Represent Chemical Reactions?

Formulae of Elements

- Metals and Group 0: single symbol — Na, Fe, Cu, He, Ar.
- Diatomic non-metals: H_2, N_2, O_2 and the halogens F_2, Cl_2, Br_2, I_2.
- Simple covalent compounds: H_2O, CO_2, CH_4, NH_3, HCl, SO_2.

Common Ions

Positive ions	Negative ions
1+: H^+, Li^+, Na^+, K^+, Ag^+, NH_4^+ (ammonium)	1–: F^-, Cl^-, Br^-, I^- (halides), OH^- (hydroxide), NO_3^- (nitrate) 2–: O^{2-} (oxide), S^{2-} (sulfide), SO_4^{2-} (sulfate), CO_3^{2-}

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2+: Mg^{2+}, Ca^{2+}, Ba^{2+}, Cu^{2+}, Fe^{2+}, Zn^{2+} 3+: Al^{3+}, Fe^{3+}	(carbonate)

Working Out Ionic Formulae

The total positive charge must **equal** the total negative charge (the compound is neutral). Use brackets for more than one compound ion.

Worked Example: Write the formulae of potassium bromide, sodium oxide, magnesium chloride and calcium hydroxide.
K^+ and Br^-: one of each → **KBr**
Na^+ and O^{2-}: need two Na^+ to balance 2– → **Na_2O**
Mg^{2+} and Cl^-: need two Cl^- → **$MgCl_2$**
Ca^{2+} and OH^-: need two OH^- → **$Ca(OH)_2$**

All Group 1 halides have the formula **MX** (1+ and 1– ions in a 1 : 1 ratio): LiF, NaCl, KBr, KI… and all Group 1 hydroxides are **MOH** (LiOH, NaOH, KOH).

State Symbols

Symbol	Meaning	Example
(s)	solid	Na(s), NaCl(s), I_2(s)
(l)	liquid	H_2O(l), Br_2(l)
(g)	gas	H_2(g), Cl_2(g)
(aq)	aqueous — dissolved in water	NaOH(aq), KBr(aq)

Writing Balanced Equations with State Symbols

Worked Example: Potassium reacts with water to form potassium hydroxide solution and hydrogen. Write a balanced symbol equation with state symbols.
Word equation: potassium + water → potassium hydroxide + hydrogen

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Unbalanced: $K + H_2O \rightarrow KOH + H_2$ (H: 2 on left, 3 on right)
Double KOH and H_2O, then K: **$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$**

Worked Example: Write the equation for bromine displacing iodine from sodium iodide solution.
$Br_2(aq) + 2NaI(aq) \rightarrow 2NaBr(aq) + I_2(aq)$ (Br: 2 = 2, Na: 2 = 2, I: 2 = 2 ✓)

Exam Tip: Never change a formula to balance an equation (e.g. writing $NaCl_2$ or H_2O_2). Halogens are always diatomic in equations (Cl_2, not Cl), but in compounds they are single ions (NaCl). Check every element on both sides at the end.

C2.5 What Are the Properties of the Transition Metals?

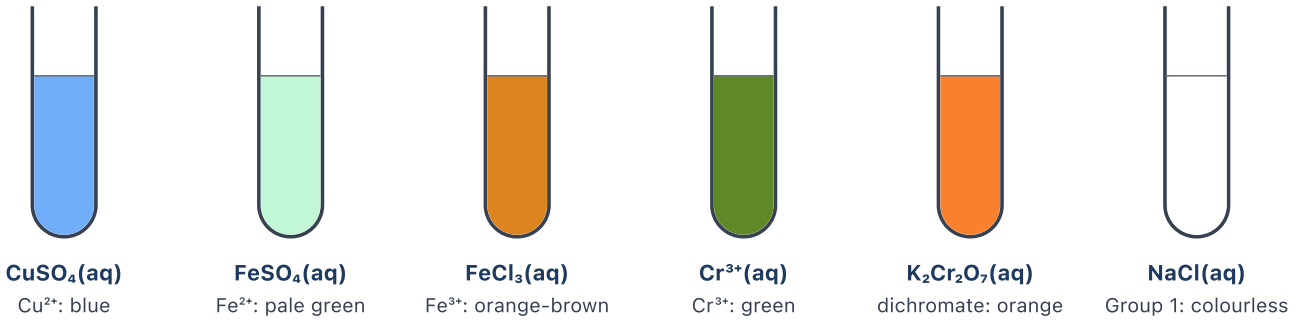
The **transition metals** are the block of metals in the middle of the Periodic Table (between Groups 2 and 3). They do not show clear group trends like Groups 1 and 7; instead they form a family with similar **general properties** that make them very useful.

General property	Transition metals	Compared with Group 1 metals
Melting point	**High** (e.g. iron 1538 °C, chromium 1907 °C, copper 1085 °C)	Low (sodium 98 °C)
Density	**High** (e.g. iron 7.9, copper 8.9, gold 19.3 g/cm³)	Low (sodium 0.97 g/cm³ — floats)
Reactivity	**Less reactive**; react slowly or not at all with water; some are very unreactive (silver, gold)	Very reactive with water and air
Ions	Form **ions with different charges** (e.g. Fe^{2+} and Fe^{3+}; Cu^{+} and Cu^{2+})	Only 1+ ions

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Colour of compounds	Form **coloured** compounds and solutions	White compounds; colourless solutions
Catalysts	Metals and their compounds are often good **catalysts**	Not used as catalysts

Transition Metal Compounds Are Coloured



Iron forms two different ions (Fe^{2+} and Fe^{3+}) with different colours

Metal	Ions and colours	Uses (linked to properties)
Copper	Cu^{2+} compounds blue (e.g. copper(II) sulfate solution)	Electrical wiring (excellent conductor, ductile), water pipes (unreactive); catalyst in some reactions
Iron	Fe^{2+} pale green; Fe^{3+} orange-brown (yellow-brown solution)	Steel for construction and cars (strong); iron is the **catalyst in the Haber process** for making ammonia
Chromium	Cr^{3+} green; dichromate ($Cr_2O_7^{2-}$) orange	Chromium plating (shiny, resists corrosion); stainless steel (with iron and nickel)
Silver	Ag^+ (colourless in solution; silver compounds such as silver halides are used in photography)	Jewellery and cutlery (shiny, unreactive), electrical contacts (best conductor)
Gold	Very unreactive — does not tarnish or corrode	Jewellery, electrical connectors in electronics, dentistry; nanoparticles of

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		gold are used as catalysts

Practical: Colours and Catalysis

- **Colours:** make solutions of transition metal compounds (e.g. copper(II) sulfate, iron(II) sulfate, iron(III) chloride, chromium(III) salts) and compare them with Group 1 compounds (e.g. sodium chloride — colourless).
- **Catalysts:** add equal small masses of different metal oxides (e.g. copper(II) oxide, iron(III) oxide, manganese(IV) oxide) to equal volumes of hydrogen peroxide solution; compare the volume of oxygen collected in a gas syringe in a set time (relights a glowing splint). The best catalyst gives oxygen fastest. Filter, dry and reweigh the solid afterwards to show the catalyst is not used up.

Ideas about Science (IaS4). Transition metals and their compounds are used widely to make consumer goods (wiring, steel, jewellery, electronics) and as industrial **catalysts** (e.g. iron in the Haber process), which save energy and resources — *beneficial applications of science*.

Exam Tip: Exam questions often give a table of data (melting point, density, colour of compounds) and ask which element is a transition metal. Look for: high melting point, high density, coloured compounds, more than one ion charge. Learn the five named examples — copper, iron, chromium, silver and gold.

Summary & Key Equations

- Atomic models: Greeks → Dalton (solid spheres) → Thomson (plum pudding) → Rutherford (nucleus) → Bohr (shells)
- Proton +1, mass 1; neutron 0, mass 1; electron −1, mass ≈ 0. Atom ≈ 10^{-10} m; nucleus ≈ 10^{-15} m
- neutrons = mass number − atomic number; electrons = protons − charge on the ion
- Group number = outer electrons; period number = number of shells

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- Mendeleev: ordered by atomic mass, left gaps, predicted properties, swapped Te/I — confirmed by discoveries and atomic numbers
- Group 1: $2M + 2H_2O \rightarrow 2MOH + H_2$; $2M + Cl_2 \rightarrow 2MCl$; reactivity increases down
- Group 7: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$; reactivity decreases down
- Group 0: full outer shell, monatomic, unreactive, low boiling points
- Ionic bonding = electron transfer + electrostatic attraction between oppositely charged ions in a giant lattice
- Ionic compounds: high mp; conduct when molten or aqueous, not solid; often soluble
- State symbols (s) (l) (g) (aq); formulae from balancing ion charges
- Transition metals: high mp and density, less reactive, coloured ions with different charges, catalysts

Specification Checklist

Ref	Can you...
C2.1.1	describe how and why the atomic model changed, including Dalton, Thomson, Rutherford and Bohr
C2.1.2	describe the atom as a small positive nucleus (most of the mass) surrounded by negative electrons
C2.1.3	recall the relative charges and approximate relative masses of protons, neutrons and electrons
C2.1.4	estimate the size and scale of atoms relative to other particles
C2.1.5	recall the typical size (order of magnitude) of atoms and small molecules
C2.1.6	relate the size and scale of atoms to everyday objects
C2.1.7	calculate numbers of protons, neutrons and electrons in atoms from atomic and mass numbers or the Periodic Table

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C2.2.1	explain how an element's position in the Periodic Table relates to its electron arrangement and atomic number
C2.2.2	describe how Mendeleev organised the elements using properties and relative atomic masses
C2.2.3	describe how new elements and ordering by atomic number supported Mendeleev's gaps and reordering
C2.2.4	describe and explain the differences between metals and non-metals (physical and chemical properties)
C2.2.5	recall Group 1 properties and reactions with moist air, water and chlorine
C2.2.6	recall Group 7 states and colours, gas colours, reactions with Group 1 metals and displacement reactions
C2.2.7	predict reactions and reactivity from position in the Periodic Table
C2.2.8	describe experiments to show the reactivity pattern of Group 7, including displacement (PAG1)
C2.2.9	describe experiments to show the reactivity pattern of Group 1
C2.3.1	recall Group 0 properties: low melting/boiling points, gases, unreactive
C2.3.2	explain how properties of Groups 1, 7 and 0 depend on the outer shell and predict properties from trends
C2.3.3	explain how reactions relate to electron arrangement and atomic number
C2.3.4	explain how the atomic structure of metals and non-metals relates to their position
C2.3.5	describe the nature and arrangement of bonds in ionic compounds
C2.3.6	explain ionic bonding by electron transfer and electrostatic forces
C2.3.7	calculate protons, neutrons and electrons in atoms and ions

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C2.3.8	construct dot and cross diagrams for simple ionic substances
C2.3.9	explain how the bulk properties of ionic materials relate to their bonding
C2.3.10	use energy transfers and the strength of attraction between ions to explain melting points compared with other bonding
C2.3.11	describe the limitations of dot and cross diagrams and 3-D representations of ions and ionic compounds
C2.3.12	translate between diagrams and numbers, and between 3-D and 2-D representations of ionic structures
C2.4.1	write formulae of elements and simple covalent and ionic compounds
C2.4.2	use the formulae of common ions to deduce formulae of Group 1 and Group 7 compounds
C2.4.3	use names and symbols of the first 20 elements, Groups 1, 7, 0 and other common elements to write formulae and balanced equations
C2.4.4	use state symbols (s), (l), (g) and (aq)
C2.5.1	recall general properties of transition metals (melting point, density, reactivity, coloured ions with different charges, catalysts) using copper, iron, chromium, silver and gold

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